

Learning Humanoid Locomotion Skills for Material Handling and Transportation in Construction Sites

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Abstract—The increasing demand for automation in construction necessitates robotic solutions that can address labor shortages, safety concerns, and efficiency challenges. Humanoid robots, with their human-like form factor, are particularly well-suited for traversing and operating in human-centric construction environments. We present HuRoCon, an integrated humanoid robotic framework designed for robust locomotion and payload transportation across diverse construction surfaces. Our system combines dynamic bipedal walking with effective payload-carrying mechanisms, enabling seamless transitions between surfaces such as flat ground, ramps, stairs, narrow beams, random obstacles, and irregular trenches while carrying loads of up to 9 lbs. We employ a unified training approach using reinforcement learning to achieve stable locomotion and real-time adaptation of walking styles based on surface conditions. Experimental results demonstrate that our framework enables continuous locomotion at speeds up to 0.42 m/s across complex construction environments, showcasing the potential of humanoid robots in construction logistics.

I. INTRODUCTION

The demand for construction automation has increased significantly in recent years due to persistent labor shortages [1], mounting concerns over worker safety [2], and the need for optimizing operational efficiency [3], [4]. The European Centre for the Development of Vocational Training (CEDEFOP) estimates that by 2025, the European Union will require approximately 1 million new and replacement workers in the construction sector [1]. At the same time, irregular working hours and long working hours can have adverse effects on the safety and health of construction workers [2]. This highlights both commercial potential and societal urgency of building automation. Robots can mitigate these issues by reducing human exposure to hazardous, repetitive, and physically demanding tasks, such as material transportation, welding operations and excavation projects [3], [5]. However, construction sites present complex, dynamic, and unstructured environments characterized by uneven surfaces, debris, and constant changes, which would require robots to handle irregular surfaces, maintain balance, and traverse narrow pathways amid moving machinery, falling objects,

etc. These factors have set extremely high requirements for the mechanical design and control strategies of construction-oriented robots.

Existing technologies, such as wheeled or tracked robots, have demonstrated significant advances in locomotion across various surface types [6]. However, these platforms often face limitations in unstructured environments. Since construction sites are predominantly designed with human-centric infrastructure [7], wheeled robots may be unsuitable for traversing narrow beams, scaffolding, or trenches with discontinuous gaps. In contrast, while quadrupedal robots have seen progress in traversing challenging surfaces [8], they lack the essential payload-carrying capabilities to handle construction materials designed for human use. Additionally, their payload capacity is generally lower compared to humanoids [9]. These observations show that mobility alone is insufficient; real deployment also requires a form factor that can lift, position, and release human-sized materials. A robot incapable of moving drywall sheets, toolboxes, or pipe sections shifts that burden back on workers, negating productivity gains. Hence, morphologies with greater anthropomorphic reach and load accommodation merit deeper investigation.

Humanoid robots exhibit significant practical potential and promising applicability in construction sites originally designed for human workers. [10], [11]. Features like stairs, ladders, and tools are tailored to human proportions, enabling humanoid robots to traverse these spaces, replicate human actions, and perform tasks such as walking on flat surfaces [12], stair climbing [13], and carrying objects [10]. Recent surveys underline that a familiar, anthropomorphic morphology helps human co-workers interpret robot intent more readily, fostering smoother collaboration and reducing onboarding effort [10]. In addition, a human-scale workspace lets the robot employ off-the-shelf tools and personal protective equipment, minimizing costly site retrofits and aligning with the principles of Industry 5.0 centered on humans [14]. On the control side, learning-based footstep planners already achieve centimetre-level accuracy on predefined paths [12], while perceptive internal model policies extend these capabilities to stairs and uneven ground without manual re-tuning [13]. Due to their potential to work alongside humans without necessitating significant modifications to existing environments or workflows, humanoid robots emerge as strong candidates for augmenting the construction workforce, offering a versatile platform for mobility and real-time material handling [15].

Nevertheless, the integration of the above-mentioned capabilities of humanoid robots for construction operations,

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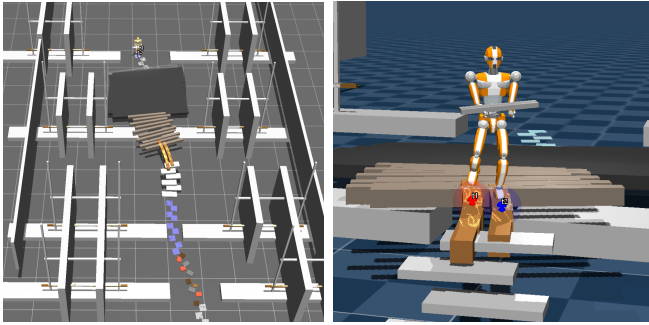


Fig. 1: LEFT: A construction site featuring diverse surface types, including flat ground, ramps, stairs, narrow beams, obstacles, etc., to represent a common construction site. RIGHT: A humanoid robot traversing the site while carrying a payload of 9 lbs.

particularly combining locomotion with carrying payloads, remains largely unexplored. The challenge lies in developing systems that can simultaneously maintain balance, adapt to varying surfaces, and carry objects— all while traversing the complex and dynamic construction environment. Achieving this integration requires a control architecture that considers foot placement, whole body momentum, and payload disturbances within a single loop. Moreover, it must execute in real time on embedded hardware and remain resilient to sensor noise, imperfect terrain estimates, and sudden external pushes typical of busy job sites.

We present an integrated humanoid robotic framework that addresses these challenges by combining dynamic bipedal locomotion with payload-carrying capabilities across diverse construction surfaces. The surfaces include plain ground, uneven surfaces, ramps, scaffolding, stairs (alternating tread and crossover), narrow beams, trenches at irregular distances, randomly placed obstacles, and slippery surfaces, enabling the robot to seamlessly transition between different surfaces while carrying construction materials. Central to our approach is a curriculum-driven reinforcement-learning (RL) pipeline that introduces terrain difficulty in stages, allowing the policy to master fundamental gaits before confronting hazardous substrates. This staged exposure produces a *surface-agnostic* controller that no longer requires manual mode switching—even when a single transport run spans flat concrete, plywood ramps, and narrow I-beams.

- **Unified RL curriculum.** We develop a unified training approach that enables robust bipedal locomotion across various challenging surfaces typical of construction sites, including narrow beams, ramps, irregular trenches, obstacles, stairs, etc.
- **Integrated whole-body control.** We propose an integrated system combining dynamic walking with payload-carrying capabilities that are essential for construction material transportation tasks.
- **On-the-fly adaptation.** It enables real-time adaptation of walking styles during transitions between different surfaces, allowing continuous locomotion within a single run while maintaining balance by following precise footsteps with payloads.
- **Validation across multiple surfaces.** We demonstrate a

surface-aware velocity control system that adjusts based on surface conditions, ensuring safe and efficient material transportation across construction sites.

Our experimental results demonstrate that the proposed framework enables stable locomotion across various construction-relevant surfaces while carrying payloads of up to 10 lbs. The system achieves continuous locomotion at speeds up to 0.42 m/s, successfully handling transitions between different surface types, including narrow beams, ramps, and irregular trenches. To the best of our knowledge, this work represents one of the first attempts to integrate dynamic bipedal locomotion with payload-carrying capabilities for construction site applications, advancing the potential of humanoid robots in construction logistics. Taken together, the results demonstrate the feasibility of deploying a single humanoid controller for end-to-end material transport. They also outline a clear pathway toward sim-to-real transfer, perception-in-the-loop footing adjustment, and collaborative operation alongside human crews on active sites.

II. RELATED WORK

Construction automation research has progressed significantly from single-purpose robots to more versatile systems. Initial efforts focused on adapting industrial robots for specific tasks, such as concrete spraying [16], bricklaying [17], and welding [18]. Although effective for their designated tasks, these systems lack the flexibility required for dynamic construction sites [19], [20]. Advancements in wheeled and tracked robots demonstrate material transport capabilities on flat surfaces. However, their inability to traverse stairs, scaffolding, or narrow beams limits their adoption in human-centric construction sites [3], [7]. Quadrupedal platforms improve surface adaptability but have limited payload-carrying and manipulation capabilities [9], motivating humanoid solutions that mirror human mobility [21].

Early single-purpose systems typically operated within fenced-off work cells or prefabricated jigs, requiring extensive site preparation and restricting throughput once layouts changed [16]–[18]. More recent mobile platforms adopt wheeled or tracked bases with pallet jacks and conveyor accessories, achieving efficient point-to-point transport on slab floors; yet empirical studies report repeated failures at temporary ramps, debris piles and doorway thresholds - features ubiquitous on renovation and green-field projects [3], [7]. Several inspection-oriented multi-robot frameworks further highlight the need for mobility over unfinished terrain but ultimately rely on aerial or static sensors rather than physical payload handling [19], [20]. This cumulative evidence underscores a persistent gap between task-specific automation and the irregular, human-centric topography of live construction sites.

Humanoid research is divided into locomotion and payload-carrying without integration. HRP-5P pioneers construction-grade payloads [22]. It can carry a large payload and perform some construction tasks, such as installing a gypsum board [23], but relied on scripted joint trajectories and overhead hoists for initial alignment, limiting versatility

once task geometry changed. Progress in bipedal locomotion and whole-body control has improved stability and adaptability [24], [25]. Contemporary work achieves stair climbing [13] and sloped surface adaptation [6] without payloads, while payload-focused systems restrict mobility to static configurations [10]. Ali et al. [15] demonstrate humanoid locomotion on unfinished floors but omit payload handling, a limitation explicitly noted in their evaluation.

Despite significant progress, existing approaches do not often address the dual demands of construction logistics: continuous transitions across complex construction surfaces and carrying construction payload during transport. Previous implementations isolate surface adaptation [13] from payload-carrying mechanisms [10]. In summary, they either secure loads at the expense of mobility (for example, stationary bolt-tightening arms [18]) or demonstrate agile locomotion while neglecting the mass, inertia, and safety constraints inherent to material transport [13], [15]. No prior study simultaneously validates seamless transitions across beams, trenches, ramps, and slippery slabs and active stabilization of a construction-relevant payload during those transitions. Therefore, meeting this dual requirement is crucial to closing the last meter automation gap between point-to-point conveyance and true end-to-end logistics at evolving construction sites. In this circumstance, our work bridges this gap through validated unified training on six types of complex surfaces with adaptation capability while maintaining payload.

III. METHODOLOGY

A. Overview

This work introduces a unified reinforcement learning (RL) framework for end-to-end training of whole-body humanoid locomotion skills tailored to material transportation tasks in construction environments. Extending prior work in bipedal gait generation, we present algorithmic and architectural innovations that enable humanoid robots to traverse complex, unstructured terrains while safely and stably transporting payloads.

We propose a construction-specific unified training architecture that allows a single RL policy to generalize across diverse and uneven surface types commonly found on construction sites, eliminating the need for manual controller switching. To address the demands of payload transport, we extend traditional locomotion frameworks by incorporating coordinated whole-body control, wherein the arms and upper body actively contribute to maintaining payload stability during dynamic movements.

Our reward function is augmented to reflect construction-specific objectives. In addition to established locomotion terms, we introduce novel reward components for payload stability, penalizing oscillations or tilting during transport, and energy efficiency, incentivizing smooth and predictable motions critical for safe operation around human workers. We further modify control parameters to optimize gait timing for construction environments, emphasizing extended stance phases that improve stability on irregular or discontinuous surfaces.

To facilitate robust skill acquisition, we design a six-stage curriculum learning pipeline that incrementally exposes the agent to increasingly challenging construction-relevant scenarios. The curriculum begins with locomotion on flat ground and systematically progresses through narrow beam walking, trench traversal, obstacle negotiation, and stair climbing. Each stage incorporates domain-specific parameterization and terrain augmentation to emulate the spatial constraints, instability, and variability inherent to real-world construction sites. Fig. 2 shows the overview of the framework.

B. Humanoid Robot and Simulation Platform

We use the JVRC-1 humanoid robot [26], which was developed for the Japan Virtual Robotics Challenge. It stands about 172 cm tall, weighs approximately 62 kg, and has 32 degrees of freedom (DoF). The JVRC-1's anthropomorphic design makes it particularly suitable for construction site applications, as its human-like proportions and kinematics enable it to navigate spaces and handle materials designed for human workers. The robot's height closely matches the average human worker, allowing access to standard construction work surfaces and scaffolding platforms without modification. Its 62 kg mass provides sufficient stability for payload carrying while maintaining the agility required for dynamic locomotion across varied surfaces. We focus on its lower-body joints (e.g., hip, knee, ankle) for learning locomotion, while arm/hand articulation supports payload-carrying tasks. The JVRC-1's joint specifications and torque limits are representative of what might be achievable in a real construction-grade humanoid robot, making our simulation results more transferable to potential real-world implementations.

All experiments are conducted in the MuJoCo physics engine [27], [28]. We choose MuJoCo for its computational efficiency and accurate contact simulation capabilities, which are crucial for modeling the complex foot-ground interactions that occur when a humanoid robot traverses diverse construction site surfaces. The engine's ability to accurately model contact physics is essential for simulating scenarios such as foot-ground interaction on slippery ramps or narrow beams, conditions fundamental to construction site navigation.

A modular construction site is employed to evaluate the robot's walking and payload-carrying skills in realistic construction environments. This modular approach allows us to systematically construct diverse training and testing environments that reflect the complexity and variability of actual construction sites. The modular design enables controlled, repeatable experiments that would be difficult, expensive, or potentially dangerous to conduct in a real-world construction setting initially.

C. RL-based Robot Behaviors

To formulate humanoid robot behavior as a Reinforcement Learning (RL) task for construction material handling, we adapt and extend the hierarchical control framework of

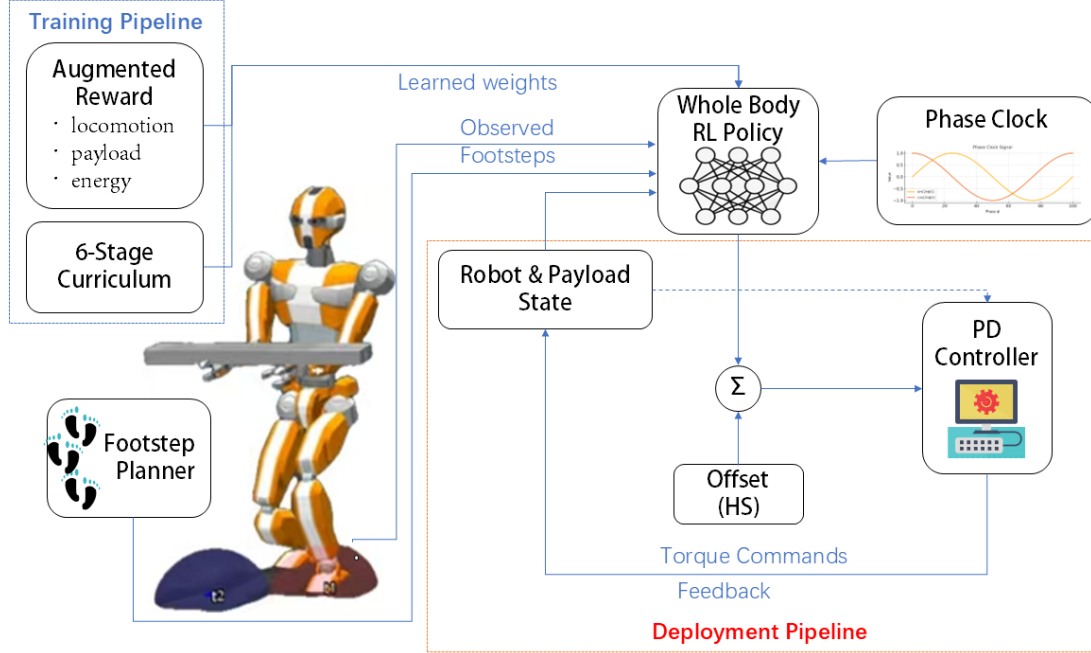


Fig. 2: Overview of the RL framework for humanoid robot locomotion in construction environments. The system has two main parts: a training pipeline that uses construction-focused rewards and a six-stage learning curriculum with footstep planning, and a deployment pipeline where the trained policy controls the robot’s whole body movements. The framework processes footstep observations and uses phase timing coordination with PD control to generate stable locomotion while carrying payloads across different construction surfaces.

Singh et al. [12], with significant modifications for payload-carrying capabilities and construction-specific locomotion requirements. Our control framework employs a two-level hierarchy consisting of a higher-level RL policy operating at 40 Hz that predicts desired joint positions, and a lower-level PD controller running at 1000 Hz that converts these position commands to joint torques. This hierarchical approach allows the RL policy to focus on high-level locomotion planning while the PD controller handles low-level tracking dynamics. The PD controller uses deliberately low gains, creating intentional tracking errors that the policy learns to exploit for generating beneficial interaction forces during contact with construction surfaces. This design is particularly advantageous for construction environments where compliant interactions with irregular surfaces are essential for maintaining stability.

The robot’s state representation is specifically designed for construction material handling tasks. Our state vector includes both leg joint positions and velocities, and critically, arm joint configurations to support payload-carrying. The state comprises joint positions and velocities of all actuated joints (legs and arms), pelvis roll and pitch angles, and pelvis angular velocities. We deliberately omit the pelvis yaw angle to define walking direction relative to the robot’s current orientation, which is suitable since footstep targets are provided in the robot’s local frame by an external footstep planner [29]. We augment the observation space with payload mass, center of mass offset indicators. This extension enables the policy to adapt its locomotion strategy based on the carried load characteristics, which is essential for safe material transport in construction sites.

For construction site navigation, we include the positions and headings of the next two footstep targets $\{T_1, T_2\}$ expressed in the local pelvic frame: $T_1 = [x_1, y_1, z_1, \theta_1]$ and $T_2 = [x_2, y_2, z_2, \theta_2]$. This two-step look-ahead is crucial for construction environments where precise foot placement on narrow beams or irregular surfaces requires anticipatory planning. We incorporate a 2D sinusoidal clock signal to maintain gait periodicity:

$$\text{Clock} = [\sin(2\pi\phi/L), \cos(2\pi\phi/L)], \quad (1)$$

where ϕ runs from 0 to L over one gait cycle. This smooth representation prevents discontinuous jumps that could destabilize learning. The policy outputs desired joint positions for all actuated joints, which are added to fixed motor offsets corresponding to a stable carrying posture before being sent to the PD controllers.

We design a phase-based reward to encourage robust bipedal locomotion and payload handling. A symmetric gait alternates between the double-support (DS) and single-support (SS) phases. Following Singh et al. [12] and Siekmann et al. [30], a gait cycle is split into two SS phases (left and right swing) and two DS phases (both feet on the ground), respectively. Following established practices, we split each gait cycle into two SS phases (left and right swing) and two DS phases (both feet on ground). For construction applications, we allocate 0.80 s for SS and 0.40 s for DS per foot, providing longer stance phases for enhanced stability when carrying payloads on irregular surfaces. Our reward function includes additional terms for payload stability, surface-adaptive stepping, and construc-

tion safety considerations, ensuring that the learned policy prioritizes material security and worker safety, which are paramount in construction environments.

1) *Periodic Bipedal Gait Reward*: To encourage proper ground reaction force during stance while minimizing force during swing, we use the periodic bipedal gait reward functions. Higher values of these rewards suggest better balance and weight transfer. This reward also encourages fast swing-foot movement while keeping the support-foot velocity low; higher scores indicate well-timed foot transitions and stable stepping. The core locomotion rewards encourage proper ground reaction force patterns and coordinated foot movements:

$$r_{\text{grf}} = I_{\text{left}}^{\text{grf}}(\phi) F_{\text{left}} + I_{\text{right}}^{\text{grf}}(\phi) F_{\text{right}}, \quad (2)$$

$$r_{\text{spd}} = I_{\text{left}}^{\text{spd}}(\phi) S_{\text{left}} + I_{\text{right}}^{\text{spd}}(\phi) S_{\text{right}}, \quad (3)$$

where $F_{\text{left/right}}$ represent normalized foot ground reaction forces, $S_{\text{left/right}}$ denote horizontal foot speeds, and $I_{\text{left/right}}^{\text{grf/spd}}(\phi)$ are phase indicator functions that smoothly transition between DS and SS intervals. Eq. 2. evaluates the ground reaction forces and Eq. 3 evaluates the foot speeds. During SS phases, the swing foot is encouraged to move efficiently while its ground contact is penalized, whereas the support foot is rewarded for maintaining contact while minimizing unnecessary motion. During DS phases, both feet are rewarded for contact forces but penalized for excessive movement. The GRF and speed rewards encourage phase-aligned contact forces and swift swings in SS, while DS stabilizes both feet on the ground.

2) *Footstep Placement and Orientation Reward*: This reward component is designed to ensure the robot not only advances toward its target but also places its feet with high precision—an essential requirement for navigating the constrained and unpredictable surfaces commonly found on construction sites. The *step reward* encourages the robot to align its foot placement with the designated target point, promoting accurate traversal over beams, gaps, or irregular terrain. It combines two elements: the *hit reward*, which is maximized when either foot lands within a specified radius of the target step T_1 , and the *progress reward*, which incentivizes the robot's root (pelvis) to move closer to the target, thereby supporting continuous forward motion. This dual focus balances the need for precise foot placement with efficient progression, both critical for safe and effective material transport in active construction environments.

The orientation reward complements this by encouraging the robot to maintain its heading toward the next target, minimizing unnecessary lateral or rotational deviations. In construction settings, where spatial constraints and the presence of obstacles are common, maintaining a consistent orientation is vital for both safety and task efficiency.

$$r_{\text{step}} = k_{\text{hit}} \exp(d_{\text{foot}}/\alpha) + (1 - k_{\text{hit}}) \exp(d_{\text{root}}/\beta), \quad (4)$$

$$r_{\text{orient}} = \exp(-6 \|\mathbf{e}_{\text{yaw}} - \hat{\mathbf{e}}_{\text{yaw}}\|^2). \quad (5)$$

In Eq. 4, d_{foot} denotes the distance from the nearest foot to the target T_1 , while d_{root} is the horizontal distance from the robot's pelvis to T_1 . We set $k_{\text{hit}} = 0.75$, $\alpha = 0.2$, and $\beta = 2.0$ to prioritize accurate foot placement while still rewarding forward progress. The orientation reward in Eq. 5 penalizes deviations from the desired yaw, scaled to ensure the robot consistently faces its intended direction, a property that is particularly important when maneuvering through cluttered construction areas.

3) *Upright Posture and Height Maintenance Reward*: Maintaining an upright posture and stable height is crucial for both the robot's own balance and the safety of transported materials, especially when traversing uneven ground or carrying heavy payloads. The upright posture reward encourages the robot to keep its torso or head vertically aligned above its base, reducing the risk of excessive leaning or swaying, which could compromise stability or lead to collisions in tight construction spaces. The height maintenance reward ensures the pelvis remains close to a nominal value, reflecting the stable gait required for reliable operation on variable surfaces.

$$r_{\text{upper}} = \exp(-10 \|\mathbf{p}_{\text{head}}^{x,y} - \mathbf{p}_{\text{root}}^{x,y}\|^2), \quad (6)$$

$$r_{\text{height}} = \exp(-30 (h_{\text{root}} - \hat{h}_{\text{root}})^2). \quad (7)$$

Here, Eq. 6 penalizes horizontal displacement between the head and pelvis, promoting a vertical upper body, while Eq. 7 penalizes deviations from the nominal pelvis height. The scale factors (10 and 30, respectively) are chosen to strongly enforce vertical alignment and height stability, both of which are essential for safe locomotion and material handling in construction environments.

4) *Payload Stability and Energy Terms Reward*: The payload stability reward ensures that the object being transported remains close to its intended position, reducing the risk of dropping or damaging construction materials during transit. This is especially important in dynamic or cluttered environments where sudden movements or impacts can destabilize the load.

Additionally, smooth and energy-efficient movements are encouraged through rewards that penalize abrupt changes in joint torques and actions. By limiting rapid fluctuations in control signals, these terms promote safer, more predictable robot behavior, which is vital for operating alongside human workers and sensitive materials on site.

$$r_{\text{payload}} = \exp(-5.0 \|\mathbf{p}_{\text{object}} - \mathbf{p}_{\text{nominal}}\|^2), \quad (8)$$

$$r_{\text{torque}} = \exp(-0.3 \|\tau - \tau_{\text{prev}}\|), \quad (9)$$

$$r_{\text{action}} = \exp(-0.2 \|a - a_{\text{prev}}\|). \quad (10)$$

In Eq. 8, a coefficient of 5.0 ensures that even small deviations from the nominal payload position are penalized, promoting secure and stable transport. The torque and action terms in Eqs. 9 and 10 (with coefficients 0.3 and 0.2) discourage sudden changes in control, leading to smoother and more reliable robot motions.

5) *Composite Reward Function*: The overall reward is computed as a weighted sum of the individual components, allowing the policy to balance the often competing demands of locomotion stability, precise stepping, payload security, orientation control, and energy efficiency. This composite approach is particularly well suited for construction applications, where operational safety, reliability, and adaptability are critical.

$$\begin{aligned} r = & w_1 r_{\text{grf}} + w_2 r_{\text{spd}} + w_3 r_{\text{step}} \\ & + w_4 r_{\text{orient}} + w_5 r_{\text{payload}} + w_6 r_{\text{height}} \\ & + w_7 r_{\text{upper}} + w_8 r_{\text{torque}} + w_9 r_{\text{action}}, \end{aligned} \quad (11)$$

The weights $w_1, \dots, w_9$ are set to 0.15, 0.15, 0.42, 0.10, 0.12, 0.70, 0.08, 0.05, and 0.12, respectively, reflecting the relative importance of each aspect for robust humanoid locomotion and material transport in construction environments. This formulation ensures the robot can maintain stable contact with the ground, accurately step and orient toward targets, securely handle payloads, and move efficiently—addressing the multifaceted demands of real-world construction sites.

D. Training Set-up and Procedure

The hierarchical control framework consists of two components: an RL policy that operates at a lower update rate (i.e., 40 Hz) to produce desired joint positions, and a high-frequency proportional-derivative (PD) controller (i.e., 1000 Hz) that converts these position commands into joint torques. The PD gains are deliberately chosen to allow observable deviations between commanded and actual positions, enabling the RL policy to exploit tracking errors and generate interaction forces that yield dynamic, agile motions [31]. Under this setting, an external planner can specify desired footstep placements or root orientations based on environmental perception, while the RL policy refines these directives by generating small joint-level position offsets from a nominal posture. This decoupling of high-level trajectory planning from low-level joint control results in robust locomotion and payload-carrying capabilities.

We train this control policy using Proximal Policy Optimization (PPO) [32], collecting on-policy data across multiple parallel environments and estimating advantages with GAE [33]. The policy and value functions are each modeled by a two-layer feed-forward network (MLP) with 256 hidden units per layer (ReLU activations). The probability ratio is clipped within $1 \pm \epsilon$ to avoid destabilizing updates, and we normalize states and rewards for consistency. We also use the LOSS method [34], which adds an auxiliary loss term to the standard PPO objective in order to enforce symmetry, further improving bipedal gait stability. Each training batch processes data from 64 rollouts, each lasting a maximum of 1000 control timesteps unless a termination criterion is met. We set the learning rate to 0.0001. Each episode is initialized with the robot in a half-sitting posture, with up to 2° random joint noise to introduce variability (following the approach in [31]). An episode terminates when one of the following conditions is met: the pelvis height drops

below 0.55 m, a self-collision is detected, or the episode reaches a horizon of 1000 control steps. The 1000-step limit is empirically chosen to ensure sufficient exploration of diverse locomotion strategies while discouraging static behavior, thereby enabling the robot to acquire robust bipedal walking and payload transport in MuJoCo [27], [28].

To expose the policy to various construction tasks, we generate footstep plans for each step in the form of a vector, $(x_k, y_k, h_k, \theta_k)$ [12], where x_k and y_k specify the horizontal target coordinates for foot placement, h_k denotes the vertical component (height), and θ_k defines the desired yaw orientation of the robot's torso. The subscript k indexes the step in the planned sequence that allows control over the robot's locomotion trajectory. In flat or gently undulating surfaces, illustrated in Fig. 3, the length of the footstep increases to promote forward efficiency while beams or narrow pathways reduce lateral spacing to keep both feet on limited support surfaces. Ramps and small stairs require incremental adjustments of step height for stable ascent, while low-friction patches encourage shorter or longer and more cautious steps. Trenches or uneven surfaces incorporate irregular step distances to test precision, and occasional sinusoidal curves reflect real construction zones that deviate from straight paths. Flat or slightly undulating surfaces permit random larger steps that boost forward efficiency, whereas narrow or slippery paths induce more cautious foot placement and short or longer step lengths. Obstacles, trenches, and curved paths appear randomly, prompting the robot to step around or over impediments [35]. Ramps and stairs require incremental changes in height, challenging the policy to coordinate ascending and descending motions. Randomly placed obstacles force the robot to step around or over hazards, further broadening its repertoire of maneuvers [35].

We combine this versatile footstep generation with a progressive curriculum of six training phases, illustrated in Fig. 3, spanning 40,000 iterations in total. Early stages include flat or gently curved surfaces that allow the robot to develop basic step patterns with moderate step lengths. Subsequent phases introduce low-friction regions, small height variations, and gradually steeper ramps that strengthen the policy's ability to recover from slips and sustain forward progress. The curriculum next requires the robot to traverse narrow beams, where lateral foot spacing shrinks to about 0.35 m and precise foot placement becomes critical. Random obstacles placed along the path add unpredictability, compelling the robot to dynamically adjust its steps. In the final stage, the robot encounters stairs with 0.15 m treads, inspired by Lee et al. [8], ensuring that it acquires the capacity to ascend or descend steps of varying heights. A difficulty factor increases at each phase boundary, refining parameters such as beam width or ramp slope. By incrementally introducing surface complexities, this training strategy prevents the policy from converging on narrow or trivial solutions and instead cultivates robust bipedal locomotion skills [36].

Through our training procedure, the humanoid robot learns to negotiate a range of simulated construction surfaces while maintaining balance and handling potential payloads. The

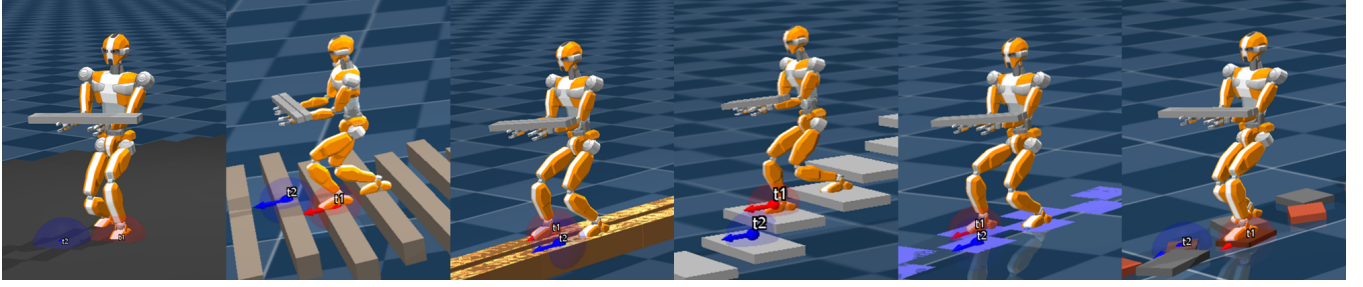


Fig. 3: Examples of the humanoid robot’s locomotion capabilities on various surfaces: Ramp, irregularly placed Trench, narrow Beam, Stairs, Slippery surface, and random Obstacle.

policy produces stable and agile motions that generalize across different scenarios encountered in training.

IV. EXPERIMENTS

We evaluate our unified reinforcement learning approach across diverse construction scenarios to demonstrate robust humanoid locomotion capabilities for material handling and transportation tasks. Our experimental evaluation follows a systematic progression: first, we analyze the robot’s performance on individual surface types, reporting velocity adaptations and locomotion characteristics. Next, we examine challenging scenarios including stair climbing, trench traversal, and narrow beam crossing. We then demonstrate adaptability in a comprehensive unified environment that integrates multiple surface transitions. Finally, we assess system robustness under realistic sensor noise and surface estimation uncertainties. These experiments collectively validate our approach’s suitability for practical material transport applications where safety and reliability are paramount.

A. Performances on Individual Surfaces

We conduct a comprehensive evaluation through 50 independent simulation runs on each surface type to assess consistency and robustness across construction-relevant terrain. This extensive testing protocol ensures statistical significance and provides confidence intervals for performance metrics essential for validating system readiness for real-world deployment. Fig. 3 illustrates the robot’s locomotion capabilities across diverse surface conditions.

Table I presents performance metrics including total distance traversed, number of steps executed, and average velocity achieved on each surface type. The results reveal significant velocity variations that demonstrate our surface-aware control system’s effectiveness. The flat surface exhibits the highest average velocity of 0.42 m/s, serving as the baseline performance under optimal conditions where the robot can maintain long, stable strides. This velocity is comparable to human walking speeds for material transport tasks.

The most challenging surface is Beam, where the average velocity drops to 0.21 m/s. This reduction reflects the need for cautious, small steps to maintain balance on the narrow pathway. Similarly, on Stairs (0.32 m/s), the robot has to negotiate discrete changes in height, which naturally reduces its speed due to the required adjustments in step length and stability. In the Slippery region, despite the low friction,

Surface	Distance (m)	Steps	Velocity (m/s)
Flat	742.5	1650	0.42
Ramp	350.0	1000	0.32
Trench	262.5	750	0.32
Beam	236.25	1050	0.21
Stairs	175.0	500	0.31
Slippery	399.0	950	0.38
Obstacles	200.0	500	0.36

TABLE I: We run the experiment 50 times for each surface. We report the number of steps, the distance covered by those steps on each surface, and the average velocity achieved on each surface. The maximum average velocity achieved is 0.42 m/s, and the lowest is 0.21 m/s. These results show that the robot can dynamically adjust its velocity and smoothly transition between different surfaces.

the robot maintains a relatively high average velocity of 0.38 m/s, indicating effective control strategies to counteract the challenging surface. Lastly, in areas with Obstacles, the robot achieves an average velocity of 0.36 m/s. It helps in balancing the need to maneuver around randomly placed obstacles with steady forward progress.

B. Performance on Stairs

Building on the stair climbing results from Table I, we provide detailed analysis of the robot’s performance on stairs with 20 cm riser height and 25 cm tread width. These dimensions are representative of standard construction access stairs and temporary stairways commonly used in building projects. The robot consistently ascends while securely carrying a 9 lb payload, representing typical construction tools or small material loads. This performance significantly surpasses existing humanoid locomotion methods, which typically demonstrate maximum climbing capabilities of 10 cm [37], [38] to 15 cm [13] without any payload burden. The maintained velocity of 0.31 m/s is particularly impressive considering the additional complexity introduced by payload management during dynamic climbing phases.

The narrow tread width of 25 cm requires precise foot placement capabilities and sophisticated balance control to ensure safe traversal without overstepping or understepping. This capability is essential for construction environments where workers frequently need to transport materials between different levels of a building under construction, validating our system’s practical applicability for multi-story construction projects.

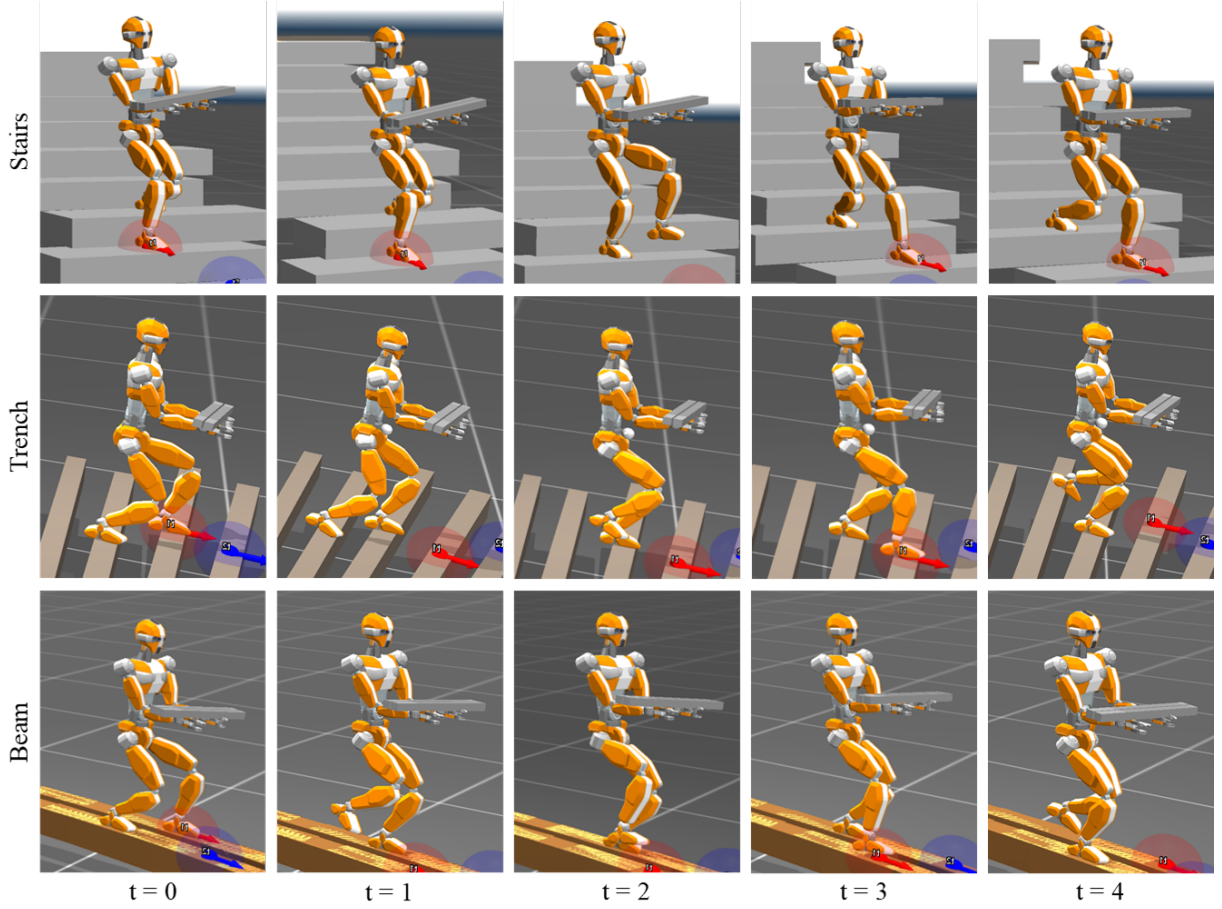


Fig. 4: TOP: JVRC-1 traverses stairs with heights of 20 cm and a width of 25 cm while carrying a payload of 9 lbs. MIDDLE: JVRC-1 successfully traverses trenches with narrow support (14 cm width) at irregular distances and with precise footsteps while carrying payloads. BOTTOM: JVRC-1 carefully balancing and crossing the narrow beam while carrying payload at a lower speed of 0.21 m/s.

C. Performance on Trench

We evaluate the robot’s traversing performance on irregularly spaced trenches supported by horizontally placed beams. These trenches range in length from 25 cm to 45 cm, creating unpredictable spacing that requires adaptive footstep planning. To test a more challenging task, the horizontal support beams measure only 14 cm in width, which is narrower than the robot’s foot. As a result, each foot placement requires high precision to avoid missteps. Fig. 4 MIDDLE shows that the robot successfully traverses this region at an average velocity of 0.32 m/s while carrying payloads. The width of these horizontal support beams exceeds typical constraints reported in existing literature [13], illustrating the robustness of the proposed approach to foot placement and balance.

D. Performance on Beam

Fig. 4 BOTTOM demonstrates the robot’s ability to traverse a narrow beam while carrying payloads. Since there exist balance constraints, the robot crosses slowly and successfully traverses a 20 cm-wide vertical beam at an average velocity of 0.21 m/s. It shows that the robot can cross a narrower beam than the current literature of 25 cm [39].

This result demonstrates the robot’s ability to transport construction materials while balancing on a beam.

E. Performance on Construction Site

To evaluate real-world applicability beyond individual surface performance, we integrate all previously described surface types into a unified construction environment. This comprehensive testing requires the robot to demonstrate seamless transitions between different surface types while maintaining continuous locomotion and payload security throughout the entire sequence.

Our experimental results, presented in Fig. 5, demonstrate that the robot achieves smooth transitions between diverse surface conditions while dynamically adjusting its velocity based on real-time terrain assessment. The velocity profile shows the practical implementation of the individual surface adaptations observed in Table I, with transitions occurring automatically without manual intervention.

The unified construction site evaluation validates our system’s core contribution: the ability to maintain continuous locomotion across diverse terrain types without requiring manual controller switching. This capability is essential for practical construction applications where surface conditions change unpredictably and frequently throughout a single

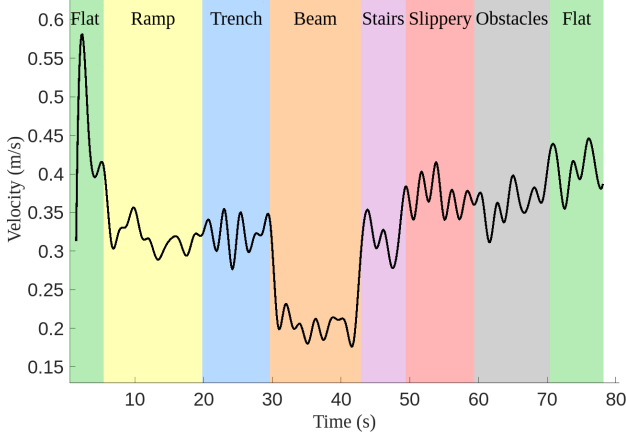


Fig. 5: Real-time velocity variations as the robot navigates diverse surfaces on a construction site. The velocity is peaked on flat terrain due to ease of movement. The initial spike represents the robot's rapid acceleration from rest. The velocity decreases significantly on beams, reflecting heightened caution. Fluctuations in velocity across trenches, slippery areas, and obstacles illustrate the robot's adaptability for safe and efficient traversing.

material transport task, extending beyond the controlled individual surface evaluations.

F. Robustness against Noise

We evaluate the robustness of our gait generation method by introducing realistic noise conditions that simulate uncertainties encountered in construction environments. Sensor noise ranging from 0° to 3° models joint encoder measurement errors and IMU drift typical in harsh construction conditions with dust, vibration, and electromagnetic interference from equipment [40], [41]. Surface noise from 0 to 3 cm accounts for inaccuracies in terrain height estimation and surface irregularities common in construction sites due to debris and material settling [42]. Both noise sources are modeled as zero-mean Gaussian distributions following established practices [43], [44].

Fig. 6 shows that under ideal conditions, the robot achieves 100% performance. With 2° sensor noise, which represents typical construction site conditions, the robot maintains near-optimal performance across all surface noise levels, with occasional drops to approximately 60% that remain acceptable for practical applications [45].

A non-monotonic response occurs at 1° sensor noise, where performance initially decreases at moderate surface noise levels before recovering at higher noise levels, suggesting complex interactions between sensor and surface uncertainties. When sensor noise reaches 3° , representing extreme conditions, average performance drops to 40–60%, indicating a critical threshold for reliable operation.

Overall, our method demonstrates robust performance with sensor noise up to 2° and across all surface noise levels, validating its suitability for real-world construction site deployment under typical operational conditions.

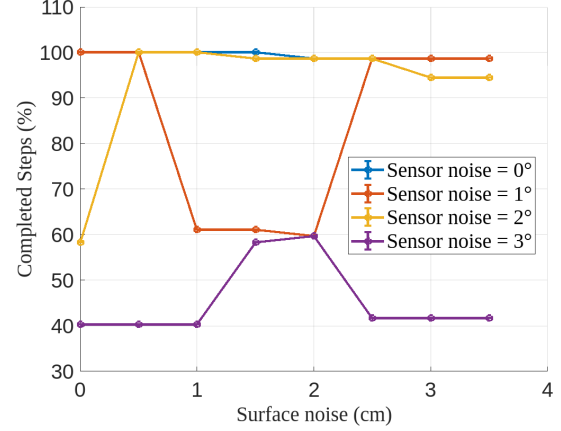


Fig. 6: Robustness evaluation showing percentage of completed steps under varying sensor noise (0° – 3° , where 0° represents ideal conditions) and surface noise (0–3 cm). Performance maintains above 60% success rate for sensor noise up to 2° , indicating strong robustness under real-world construction conditions [45].

V. CONCLUSION AND FUTURE WORK

We present HuRoCon, an integrated framework for humanoid robots to traverse construction environments while carrying payloads. Our approach demonstrates stable locomotion across various complex construction surfaces, achieving speeds ranging from 0.21 m/s on narrow beams to 0.42 m/s on flat surfaces while maintaining balance with payloads up to 9 lbs. Comprehensive experiments show that the robot achieves consistent stability and surface adaptation. HuRoCon also shows robustness against sensor noise and surface irregularities that suggest potential for practical construction tasks. Overall, our results validate the effectiveness of HuRoCon and lay a foundation for continuing deployment of humanoid robots in construction environments.

Several future directions exist. First, integrating the RL policy with an external footstep planner with additional sensors (e.g., LiDAR) would enhance the perception of the construction environment and footstep planning. Second, extending the framework to object detection and manipulation capabilities would improve the robot's construction utility. Finally, incorporating human-robot collaboration features would allow the robot to understand and respond to human actions in a shared workspace.

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